

TB Delay

Version: 1.0.0 | Documentation date: 2026-08-04

Overview

Creates repeating echoes, rhythmic space, and evolving feedback textures.

What this plug-in solves

A conventional delay repeats a fixed copy. TB Delay is designed for repeats that evolve on every cycle, while still covering precise utility echo, slap, dual, ping-pong, ratio, and tempo-locked work. Character, color, modulation, and grain transforms can live inside the feedback loop or after it.

What you can expect

- Clean utility and tempo-synced delay
- Stereo dual, ping-pong, ratio, and multi-tap patterns
- Tape, BBD, and digital character
- Pitch, frequency-shift, reverse, diffusion, and granular evolution

How it works

TB Delay stores the incoming sound and plays it back later as one or more repeats. Feedback returns those repeats to the delay so they continue, while stereo style, spacing, and timing determine whether the result feels like a simple echo, a rhythmic pattern, or a wide moving sequence.

Character, filtering, modulation, diffusion, pitch, frequency shifting, reverse, and grain processing can reshape the repeats. Transform Route decides whether those changes accumulate inside the feedback loop or remain outside it. Build a stable echo first, keep feedback controlled, then add one transform at a time so the evolution remains intentional.

Quick start

- 1 Choose Free, Sync, or Tune and set Delay Time or Division.
- 2 Choose Style, Taps, Spacing, and Width.
- 3 Set Feedback below self-oscillation first.
- 4 Choose Character and set Drive, Age, Low Cut, and High Cut.
- 5 Add modulation or Smear/Size.

- 6** Choose Loop or Post transform routing, then add Pitch, Shift, Reverse, or Grain.
- 7** Set Duck, Mix, and Output.
- 8** Use Freeze only after loop level is controlled.

Interface and key sections



This is a direct capture of the current plug-in interface. Use the visible labels to locate the areas and controls explained below.

Time modes

Free uses Delay Time in milliseconds. Sync uses host tempo and Division. Tune treats delay as a pitch-related comb interval. Time Change selects Crossfade or Glide when timing moves.

Styles and taps

Single, Dual, Ping Pong, and Ratio determine stereo topology. Taps selects 1, 2, 4, or 8 repeats; Spacing, Swing, Offset, Diverge, and Width distribute them.

Feedback and safety

Feedback extends regeneration to self-sustaining territory. Freeze holds the current loop. Change high-feedback and Freeze states carefully because internal energy can accumulate.

Character

Clean, Tape, BBD, and Digital models combine with Drive and Age. Low Cut and High Cut control feedback bandwidth.

Modulation and diffusion

Mod Rate and Mod Depth move delay time. Smear and Size diffuse repeats from discrete echoes toward a cloud.

Spectral transforms

Pitch shifts repeats in semitones, Shift frequency-shifts them in hertz, and Reverse sets reverse probability. These can create ascending, descending, or inharmonic cycles.

Granular system

Grain, Grain Size, Density, Spray, and related transform controls fragment the repeats. Grain Transform Bypass disables that transform block without changing the whole delay.

Transform Route

Loop places transforms inside regeneration so every cycle evolves; Post applies them after the feedback loop for a more stable echo core.

Ducking, Mix, and Output

Duck lowers wet activity around the dry source. Mix sets final dry/wet balance and Output trims the complete result. Stereo Bypass and Color Space Bypass isolate defined subpaths in the plug-in.

Programs

The extensive program bank covers utility delay, vocal pocket, guitars, ping-pong, golden ratio, slap families, doubles, chorus drift, reverse tails, and octave sparks.

Controllable properties

The guide uses the names shown on the plug-in panel. Each explanation describes the audible result, a useful way to approach the control, and an important interaction to listen for.

Control	What it does and when to use it
Age	Makes repeats sound newer and cleaner or older, softer, and more worn. Move it gradually and listen for the point where the intended result becomes clear without introducing a new problem elsewhere.
Bypass	Turns the complete effect off or on for a direct before-and-after comparison. Compare with bypass at a similar loudness. If the effect creates a tail, let that tail settle before judging the difference.
Character	Chooses the repeat tone: clean, tape, bucket-brigade, or digital. Compare the available characters on the same short passage. Choose by the musical result rather than by the name of the mode.
Color Space Bypass	Turns the tone and space section off while keeping the repeat timing active. Sweep broadly to find the useful tonal area, then make the focus narrower and the change smaller while listening in the full mix.
Delay Time	Sets the time between the original sound and the repeat when timing is free. Judge this control while the source repeats in the full arrangement. The right timing should support the groove without making the sound feel detached.
Density	Sets how many granular fragments overlap inside the transformed repeats. Start with a small amount, then increase it until the motion is clear without hiding the timing or identity of the original sound.
Diverge	Makes the left and right repeat paths evolve differently over time. Start with a small amount, then increase it until the motion is clear without hiding the timing or identity of the original sound.
Division	Chooses the note value used when the delay follows the song tempo. Move it gradually and listen for the point where the intended result becomes clear without introducing a new problem elsewhere.
Drive	Sets how strongly the sound is pushed into harmonic color or distortion. Level-match the result and listen for lost transients or harsh buildup. More color is useful only while the source still keeps its role in the mix.

Control	What it does and when to use it
Duck	Lowers the echoes while the dry sound is present, then lets them rise in the gaps. Move it gradually and listen for the point where the intended result becomes clear without introducing a new problem elsewhere.
Feedback	Returns the processed sound into the effect to create a longer or stronger result. Increase it slowly and watch the output level. If the sound keeps growing after the input stops, reduce this control before adding more processing.
Freeze	Holds the current effect tail so it continues without new input. Increase it slowly and watch the output level. If the sound keeps growing after the input stops, reduce this control before adding more processing.
Grain	Adds granular breakup and particle-like texture to the repeats. Start with a small amount, then increase it until the motion is clear without hiding the timing or identity of the original sound.
Grain Size	Sets the length of each granular fragment. Short values feel gritty; long values feel smoother. Start with a small amount, then increase it until the motion is clear without hiding the timing or identity of the original sound.
Grain Transform Bypass	Turns the grain and pitch transformation section off while keeping the basic delay active. Start with a small amount, then increase it until the motion is clear without hiding the timing or identity of the original sound.
High Cut	Removes high frequencies from the current sound or effect layer. Sweep broadly to find the useful tonal area, then make the focus narrower and the change smaller while listening in the full mix.
Low Cut	Removes low frequencies from the current sound or effect layer. Sweep broadly to find the useful tonal area, then make the focus narrower and the change smaller while listening in the full mix.
Mix	Blends the original sound with the processed sound. Turn the processed sound up temporarily to learn its character, then return to the mix and keep only the amount that supports the source.
Mod Depth	Sets how strongly the named process changes the sound. Set the speed first and then add only enough movement to hear the pattern without pulling attention away from the performance.

Control	What it does and when to use it
Mod Rate	Sets how quickly the named movement repeats. Set the speed first and then add only enough movement to hear the pattern without pulling attention away from the performance.
Offset	Moves one side earlier and the other later to create stereo separation. Move it gradually and listen for the point where the intended result becomes clear without introducing a new problem elsewhere.
Output	Sets the final listening level after processing. Match the final loudness before deciding whether the processing sounds better. Extra level can make almost any setting seem more impressive.
Pitch	Moves the pitch of the processed sound up or down. Choose the musical direction first, then check attacks and sustained notes for unwanted wobble or a change in character.
Program	Loads a factory starting point. Adjust the controls afterward to fit the current sound. Use a preset as a direction, not a finished answer. Change one section at a time so it is clear which adjustment improves the source.
Reverse	Sets how often or how strongly the processed sound plays backward. Start with a small amount, then increase it until the motion is clear without hiding the timing or identity of the original sound.
Shift	Moves the frequency content of each repeat upward or downward for spiral-like motion. Choose the musical direction first, then check attacks and sustained notes for unwanted wobble or a change in character.
Size	Sets the apparent size of the diffuse space around the repeats. Move it gradually and listen for the point where the intended result becomes clear without introducing a new problem elsewhere.
Smear	Softens the edges of the repeats so they blend into a more reverb-like cloud. Move it gradually and listen for the point where the intended result becomes clear without introducing a new problem elsewhere.
Spacing	Expands or contracts the distance between multiple taps. Move it gradually and listen for the point where the intended result becomes clear without introducing a new problem elsewhere.

Control	What it does and when to use it
Spray	Spreads small pieces of each repeat in time for a looser, less predictable texture. Start with a small amount, then increase it until the motion is clear without hiding the timing or identity of the original sound.
Stereo Bypass	Keeps the delay movement centered when enabled, useful for mono-safe repeats. Move it gradually and listen for the point where the intended result becomes clear without introducing a new problem elsewhere.
Style	Chooses the repeat layout, including single, dual, ping-pong, and ratio patterns. Compare the available characters on the same short passage. Choose by the musical result rather than by the name of the mode.
Swing	Moves alternating repeats earlier or later to create a looser groove. Move it gradually and listen for the point where the intended result becomes clear without introducing a new problem elsewhere.
Taps	Chooses how many repeat points are created for each echo cycle. Move it gradually and listen for the point where the intended result becomes clear without introducing a new problem elsewhere.
Time Change	Chooses whether delay-time changes crossfade cleanly or glide with a pitch bend. Judge this control while the source repeats in the full arrangement. The right timing should support the groove without making the sound feel detached.
Time Mode	Chooses free milliseconds, song-tempo divisions, or tuned pitch intervals for the delay time. Compare the available characters on the same short passage. Choose by the musical result rather than by the name of the mode.
Transform Route	Places the creative transformations inside the feedback loop or after it. Raise it until the contribution is obvious, then reduce it slightly and recheck the sound in the complete mix.
Width	Sets the stereo spread of the processed sound. Check the result on speakers, headphones, and in mono when possible. Extreme spatial settings can feel exciting but may not translate everywhere.

Practical uses

Vocal dotted eighth

- Choose Sync and 1/8 D.
- Use Ping Pong or Dual.
- High-pass and low-pass the feedback.
- Add Duck so words remain clear.
- Keep transforms off or Post for a stable repeat.

Expected result: A wide rhythmic throw supports the vocal without masking each phrase.

Evolving pitch spiral

- Use moderate Feedback.
- Set Transform Route to Loop.
- Add +7 or +12 Pitch and some Smear.
- Use High Cut and Output to contain accumulating brightness.

Expected result: Every repeat climbs and diffuses into a controlled spiral.

Granular freeze cloud

- Set a stable loop first.
- Raise Grain, Density, and Spray.
- Use Freeze only after the level is safe.
- Automate Mix and Output for entry and exit.

Expected result: The delay becomes a suspended textured cloud.

Common pitfalls

Very high Feedback and Freeze can create a runaway level. Reduce the main amount, feedback, drive, or input first, then rebuild the sound while watching the output meter and listening after the source stops.

Tempo sync requires valid host BPM. Return the strongest control toward neutral, compare with bypass at a similar loudness, and add the processing back one section at a time.

Pitch, grain, diffusion, and high tap counts increase CPU and may add artifacts intentionally. Use the lighter quality setting while recording and reserve the heavier option for playback or export. Change latency-related modes while playback is stopped.

The current host text round-trip for one delay parameter has a minor precision mismatch; audio operation is unaffected. Return the strongest control toward neutral, compare with bypass at a similar loudness, and add the processing back one section at a time.